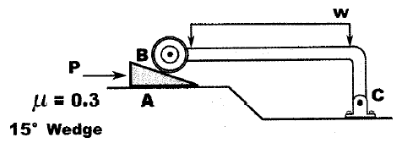
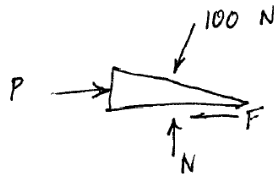
Wedge Problems from Old Exams

5. A 15° wedge is used to raise and lower the L shaped frame BC loaded with a distributed load. If frame BC exerts a 100 N force normal to the wedge at B, determine the force P to be applied to the wedge in order to initiate raising the frame at B. The coefficient of friction between the wedge and surface A is $\mu = 0.3$.



A. 54.9 N ✓
B. 22.8 N
C. 13.2 N
D. 67.4 N
E. 38.3 N



$\uparrow +$ $N = 100 \cos 15 = 96.59$

$F = .3N = 28.98$

$\overset{+}{\rightarrow}$ $P - 100 \sin 15 - 28.98 = 0$

$\boxed{P = 54.9 \text{ N}}$

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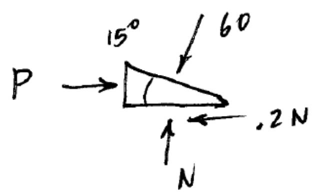
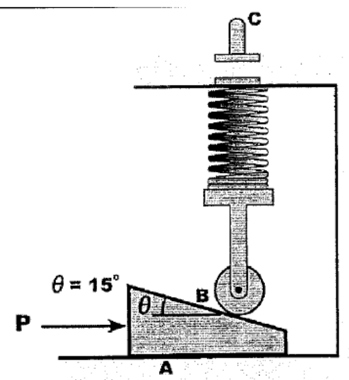
18. (5 pt) A wedge is used to raise and lower the plunger BC in the mechanism shown. At the position shown, the smooth roller exerts a 60 lb normal force onto the top of the wedge at B. Determine the P needed to initiate raising the plunger BC.

A. 17.4 lb
B. 60 lb
C. 33.2 lb
✓D. 27.1 lb
E. 11.6 lb





$\mu = 0.2$ at A
Wedge A is massless.

$\overset{+}{\rightarrow}$

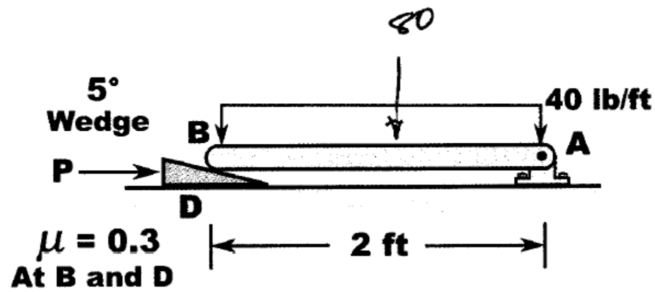
$P - 11.59 - 60 \sin 15 = 0$

$P = 27.1$ lb

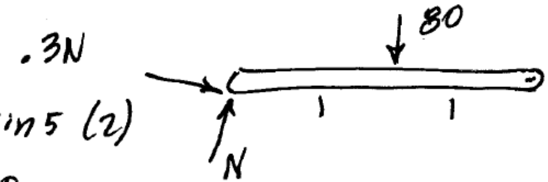
$N = 60 \cos 15 = 58$

$F = .2N = 11.59$

4. Beam BA supports a 40 lb/ft uniform load. Wedge BD is used to level the beam as shown, with force P pushing the wedge. The coefficient of friction is $\mu = 0.3$ both at the top (at B) and on the bottom (D) of the wedge. Determine the normal force at B between the beam and the top of the wedge.



A. 41.2 lb
B. 40 lb
C. 20 lb
D. 31.6 lb
E. 44.5 lb



$$\overset{+}{\curvearrowleft} \; 80(1) + .3N \sin 5\,(2) - N(\cos 5)(2) = 0$$

$$80 = (1.992 - .05229)N \Rightarrow$$

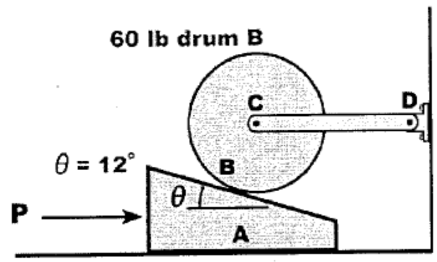
$$\boxed{N = 41.2 \text{ lb}}$$

5. To test if a wedge is self-locking, one does the following:
   A. Apply force P pulling on the wedge with P = 0.
   B Apply force P pushing on the wedge with P = 0.
   C. Apply force P pushing on the wedge with P = 0.
   D. Apply force P pushing on the wedge with $\mu = 0$.
   E. Apply force P pulling on the wedge with $\mu = 0.5$.

6. (5 pt) To test if a wedge is self-locking, one does the following:

A. Apply force P pushing on the wedge, solve for P, and for self-locking P must be ≥ 0.
B. Apply force P pulling on the wedge, solve for P, and for self-locking P must be < 0.
✓C. Apply force P pulling on the wedge, solve for P, and for self-locking P must be ≥ 0.
D. Apply force P pushing on the wedge, solve for P, and for self-locking P must be > 0.
E. Apply force P pulling on the wedge with $\mu = 0.5$.

7. (5 pt) A 12° wedge is used to raise and lower the 60 lb drum B. If the drum exerts a 61.3 lb normal force at B onto the wedge, determine the force P to be applied to the wedge in order to initiate raising the drum. The drum is smooth, and the coefficient of friction between the wedge and surface A is $\mu = 0.2$. The wedge A and the link CD are massless.

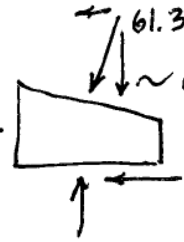


✓A. 24.7 lb
B. 20.8 lb
C. 13.4 lb
D. 38.4 lb
E. 29.2 lb

$61.3 \sin 12 = 12.74$ lb

61.3 lb

$61.3 \cos 12 = 60$ lb



P

$F = .2(60) = 12$ lb

$N = 60$ lb

$\xrightarrow{+}$ $P = 12.74 + 12 = 24.7$ lb